Pipeline before backbone: 16-hour $PM_{2.5}$ forecasting over India with Fourier neural operators and convolutional-recurrent networks

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***Abstract*: We forecast $PM_{2.5}$ over all of India 16 hours ahead, on a 140×124 grid of about 25 km cells, from 10 hours of WRF-Chem simulation fields. Phase 1 forecast one Beijing station with gradient boosting. Phase 2 uses deep learning and asks how much of a forecaster's skill comes from its network and how much from the pipeline around it. Over seven development iterations we built a physics-informed pipeline: ventilation, advection and tendency channels, a log-space persistence residual, a leak-free temporal validation, a loss matched to the evaluation metric, and seed ensembles with per-hour calibration. Five networks were then trained inside this one pipeline, with decision rules fixed before each stage and 1, 2 and 5 seeds. A U-Net with a convolutional GRU over lead time (ResGRU-UNet, 7.1M parameters) beats an adaptive Fourier neural operator hybrid (AFNO V4, 7.0M) on all 25 seed pairs and at every lead hour. On 214 validation windows its five-seed mean reaches MAE 6.32 $\mu g\,m^{-3}$, RMSE 18.04 and R^2 0.892, against 6.99, 19.45 and 0.874 for AFNO V4, 8.51, 21.48 and 0.847 for ridge regression, and 10.44, 26.46 and 0.767 for persistence: 31.8% lower RMSE than persistence. On the held-out test set it scores 0.8914 against 0.8848. Moving the same network into the improved pipeline gained 6 to 8 times more than swapping the network. Four single-change ablations do not explain the gap between the networks; training curves show that AFNO V4 generalizes less well to unseen months.**

I. Introduction

Fine particulate matter ($PM_{2.5}$) is the largest environmental health risk in India: air pollution was linked to an estimated 1.67 million deaths there in 2019 [3]. Country-wide forecasts usually come from chemistry transport models such as WRF-Chem [4], which couple weather, emissions, chemistry and transport and are expensive to run. A deep network that learns to step such a model forward from its recent fields could forecast at a small fraction of the cost.

Phase 1 [1] forecast $PM_{2.5}$ six hours ahead at one Beijing station with gradient boosting, and planned deep models for Phase 2. Phase 2 moves to a larger problem: 17,360 grid cells, 16 lead hours, and weather fields that carry pollution across the map. Two families of deep networks suit gridded forecasting. Fourier neural operators mix the whole map at once in the frequency domain [7], [8]. Convolutional-recurrent networks pass local information step by step [11], [13]. Published comparisons between them usually change the inputs, validation, loss and post-processing along with the network, so a difference in score cannot be credited to the network alone.

Our contributions are:

- a physics-informed, leak-free pipeline built over seven iterations, with the reason for each move and its measured effect (Sections V to IX);
- a controlled study of five networks inside that one pipeline, with decision rules fixed before each stage and 1, 2 and 5 seeds (Section X);
- MAE, RMSE and R^2 for every model against persistence, seasonal-naive and ridge baselines, maps of where the errors fall, and the finding that model rankings depend on the metric (Section XI);
- ablations and training curves that test why the convolutional-recurrent network wins (Section XII).

II. Related work

Chemistry transport and emulation. WRF-Chem runs weather and chemistry together on one grid with one transport scheme [4]. For particulate matter over the Indian subcontinent, lightweight deep networks were reported to beat the Aurora foundation model at short horizons [5]. CoNOAir, a complex neural operator, is reported to beat the Fourier neural operator for carbon monoxide in Indian cities [6].

Neural operators. The Fourier neural operator (FNO) learns maps between function spaces with convolutions in Fourier space [7]. The adaptive FNO (AFNO) replaces FNO's weight per mode with a small shared MLP over blocks of channels and a soft threshold that keeps the spectrum sparse [8]. AFNO is the backbone of the FourCastNet global weather model [9], and AdaptFNO adds multiscale learning for climate fields [10].

Convolutional-recurrent forecasting. ConvLSTM replaced the matrix products of an LSTM with convolutions for precipitation nowcasting [11]; TrajGRU followed [12], and the convolutional GRU is the gated-recurrent version [13]. U-Net's skip connections pass fine detail from encoder to decoder [14], and residual learning makes deep stacks easy to train [15]. SimVP showed that a purely convolutional encoder, translator and decoder can match recurrent models [16]. Shortcut recurrent-convolutional U-Nets were used for weather prediction [17].

Training and evaluation. We use AdamW [18], the one-cycle schedule [19], GroupNorm [20], weight averaging [21] and deep ensembles [22]. SMAPE is not symmetric in additive error [23]; dependent data need blocked rather than random validation [24], and information the model would not have at

TABLE I
WHAT THE DATA SHOWED (TRAINING MONTHS) AND WHAT IT CHANGED

Finding	What the model gets
Right-skewed: December mean 59.1, median 23.7 $\mu g\,m^{-3}$	Work in $\log(1+y)$
Lag-1 autocorrelation of $\log(1+y)$ is 0.978; the standard deviation of its change grows from 0.101 at 1 h to 0.505 at 16 h	Predict the change from persistence; weight later hours more
Mean 10-m wind 4.24 m/s: $PM_{2\cdot5}$ moves about 10 cells in 16 h	Advection channel; a local receptive field suffices
Strong daily cycle (correlation 0.571 at 24 h); test windows carry no clock	Estimated hour; diurnal anomaly
Hotspots over the Indo-Gangetic Plain and the eastern coal belt; monthly means 15.4 (July) to 59.1 (December)	Per-cell normalization; static climatology
Episodes are about 2% of cell-hours and are defined on the truth	Episode-weighted loss; calibration
Emission fields are tiny (10^{-8} to 10^{-7})	Not used

forecast time inflates scores [25]. Persistence and seasonal-naive are the standard forecasts to beat [26], and ridge regression [27] is our classical model.

III. Data and what the exploration showed

The data are WRF-Chem simulation fields released by the organizers of the ANRF AISEHack 2026 challenge (Theme 2, IIT Delhi) [2]. Training covers four months of 2016, one per season (April, July, October and December; 715 or 739 hours each), on a Mercator grid of 140×124 cells from 7.06 to 38.00°N and 67.79 to 97.93°E, about 25 km apart. Each hour has 16 fields: surface $PM_{2\cdot5}$ (the target), humidity, 2-m temperature, 10-m wind (u, v), downward shortwave radiation, boundary-layer height (PBLH), surface pressure, rain, and seven emission fields. The test set is 218 windows of 10 hours from 2017; its answers are hidden and scored by the organizers' evaluation server. Table I lists what the data showed and what each finding changed.

Episodes. For each cell, a robust STL decomposition [28] with a 24-hour period splits the month into trend, daily cycle and residual. An hour is an episode when its residual exceeds three standard deviations of that cell's residuals and its value exceeds 1 $\mu g\,m^{-3}$.

IV. Setup and fair testing

A. Task

Given the fields $\mathbf{X}_{t-9:t}$ of the last 10 hours, one network forecasts all 16 hours at once,

$$\hat{\mathbf{y}}_{t+h} = f_\theta(\mathbf{X}_{t-9:t})_h, \qquad h = 1, \ldots, 16, \tag{1}$$

where each $\hat{\mathbf{y}}_{t+h}$ is a 140×124 map of $PM_{2\cdot5}$ in $\mu g\,m^{-3}$. This is regression.

B. Validation without leaks

A window spans 26 hours (10 in, 16 out). Two windows whose starts differ by s hours share $(26 - s)/26$ of their hours, so with a random split almost every validation window has a training neighbour that shares 24 or 25 of its hours, including most of its answers [24]. Our early iterations used such a split and their validation scores were inflated. We hold out the last 18% of every month, leave a 12-hour gap before it, and validate on 214 windows (stride 2); training uses the 2,257 earlier windows. Every model is scored on the same 214 windows.

C. Scores

With $\mathrm{SMAPE}(p, y) = |p - y|/\big(\frac{1}{2}(|p| + |y|)\big)$, let G be the mean SMAPE over all cells and hours, E its mean over episode cells, and ρ the Pearson correlation of forecast and truth over episode cells, per hour and averaged. The benchmark score is

$$S = w_1\big(1 - \tfrac{G}{2}\big) + w_2\,\tfrac{1+\rho}{2} + w_3\big(1 - \tfrac{E}{2}\big). \tag{2}$$

The weights are not disclosed, so on validation we use $w_i = 1/3$. S is a composite of relative error and correlation, not an accuracy. With errors $e_i = \hat{y}_i - y_i$ over all $n = 5.9 \times 10^7$ validation values,

$$\mathrm{MAE} = \tfrac{1}{n}\textstyle\sum_i |e_i|, \qquad \mathrm{RMSE} = \sqrt{\tfrac{1}{n}\textstyle\sum_i e_i^2}, \tag{3}$$

$$R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i (y_i - \bar{y})^2}, \qquad \mathrm{skill} = 1 - \frac{\mathrm{RMSE}}{\mathrm{RMSE}_{\mathrm{persistence}}}, \tag{4}$$

as in Phase 1 [1]. MAE, RMSE and R^2 exist only for validation, because the test answers are hidden; the test set gives only S. Note that ρ^2 is not R^2: ρ ignores bias and scale.

D. Decision rules and checks

Before each stage of the study we wrote down how the result would be read. A network is called better only if it leads by more than 0.003 on validation and also scores higher on the test set; a smaller gap is a tie, and disagreement is inconclusive. Every training run and every main analysis was first run on synthetic data on a CPU, and every analysis recomputes the earlier numbers it depends on: for example, the regression metrics recompute each model's validation score and match its logged value within 10^{-4}.

V. Theory behind the design

Why log space. For $p, y > 0$ write $u = \log(p/y)$. Then

$$\mathrm{SMAPE}(p, y) = \frac{|p - y|}{(p + y)/2} = 2\,\big|\tanh(u/2)\big|, \tag{5}$$

since $(p - y)/(p + y) = (e^u - 1)/(e^u + 1)$. SMAPE is therefore symmetric in the log ratio: twice too high and half too low cost the same, 0.667. Because SMAPE grows with $|u|$, the median minimizes the expected global SMAPE whenever $\log y$ is symmetric and unimodal around it. A network that predicts in $\log(1 + y)$ works in the geometry the metric uses, and the log also tames the heavy right tail.

Why a persistence residual. Every network outputs a change Δ_h added to the last observed hour,

$$\hat{y}_{t+h} = \exp\big(\log(1 + y_t) + \Delta_h\big) - 1, \tag{6}$$

with the last layer set to zero at the start [15]. An untrained network is therefore exactly persistence, which already scores 0.8515 on validation, and training only has to learn the departure from it.

TABLE II
COST AND REACH PER LAYER (N CELLS, d CHANNELS, b BLOCKS, N_f FOURIER MODES)

Operator	Cost (multiply-adds)	Reach per layer
Self-attention	$2N^2d$	whole map
AFNO block	$\approx 2.5\,dN\log_2 N + 8N_f d^2/b$	whole map
Convolution $k \times k$	$Nk^2 d_{\text{in}} d_{\text{out}}$	$+(k-1)/2$ cells
ConvLSTM step	$Nk^2(d_x + d_h)\,4d_h$	$+(k-1)/2$ cells

Why the episode terms reward upward shifts. Episode cells are picked from the truth: where y turned out high against its usual level. On those cells the truth is on average 50% above persistence (mean $\log(y/\text{persistence}) = +0.406$, against -0.012 over all cells). A forecast that cannot see the future noise therefore tends to score better on E when it is shifted up; for persistence, the scale that maximizes the score rises with lead time from 1.00 to 1.30. Pearson ρ does not change when a forecast is scaled, so a per-hour factor $c_h \geq 1$ moves only the SMAPE terms. This is the basis of the calibration in Section VIII.

How far must a forecast see. At the mean wind of 4.24 m/s, $PM_{2.5}$ travels 244 km in 16 hours: 9.8 cells, or 11.5 cells at the RMS wind. Table II compares the reach and cost of the main operators for N cells, d channels and kernel size k. A 3×3 convolution reaches one more cell per layer: ResGRU-UNet's encoder alone reaches about 14 cells, and each ConvGRU step at 18×16 adds about 8 more, so it covers the needed radius. AFNO mixes the whole map in every block, at a cost close to linear in N, but transport over 16 hours does not need global reach.

VI. PHYSICS-INFORMED INPUTS

Surface $PM_{2.5}$ changes through emission, transport, vertical mixing, chemistry and deposition. Three of these leave a mark in the available fields, and we compute each one as an input instead of leaving the network to find it. Table III lists the 20 channels.

Vertical mixing. The boundary layer dilutes surface pollution in proportion to its depth and the wind that flushes it. Their product, the ventilation coefficient, measures how easily the air clears; its long decline over Indian cities goes with rising pollution potential [29]. We add

$$V = \log\bigl(1 + \text{PBLH} \cdot \|\mathbf{u}_{10}\|\bigr) \tag{7}$$

and the wind speed $\|\mathbf{u}_{10}\|$. Low V is stagnant air, where episodes build.

Transport. To first order a tracer follows $\partial_t C = -\mathbf{u} \cdot \nabla C$, so we add the advection of the log concentration,

$$A = -\bigl(u\,\partial_x + v\,\partial_y\bigr)\log(1 + C), \tag{8}$$

from centred differences. Earlier models used the same physics in other forms: ConvLSTM shifted the $PM_{2.5}$ map along the wind (a semi-Lagrangian step [30]), and FMG-WACLU added wind direction and divergence, since converging wind ($\nabla \cdot \mathbf{u} < 0$) traps pollution.

TABLE III
THE 20 INPUT CHANNELS IN FOUR GROUPS

Group	n	Channels
Weather	9	$\log(1+PM_{2.5})$, humidity, temperature, u_{10}, v_{10}, PBLH, pressure, shortwave radiation, $\log(1+$ rain)
Physics	3	wind speed, ventilation V, advection A
Time	4	tendency, hour sin and cos, diurnal anomaly
Static	4	latitude, longitude, cell $PM_{2.5}$ climatology, cell STL residual spread

TABLE IV
THE FIVE NETWORKS (STUDY VERSIONS, 20 INPUT CHANNELS)

Network	Params	Design and inductive bias
ResGRU-UNet	7.12M	all 10 hours stacked as 200 channels; conv U-Net; ConvGRU unrolled over the 16 lead hours
AFNO V4	6.98M	per-hour conv encoder; AFNO blocks at two scales; temporal translator; U-Net decoder
FMG-WACLU	3.45M	ConvLSTM U-Net with wind physics and a gate from forecast weather
patch AFNO	5.71M	4×4 patches, AFNO blocks, linear unpatchify
ConvLSTM	19.0M	ConvLSTM over the inputs, spectral block, wind-shift advection

Time. A window-local tendency $\Delta_t \log(1 + C)$, divided by its per-cell spread, separates building from clearing episodes. The test windows carry no clock, so each window's start hour is estimated by matching its mean sunlight and temperature to hourly templates from training; this is right within one hour for 97.7% of validation windows, and validation uses the estimate exactly as the test does. The hour enters as $(\sin, \cos)$, with the departure of $\log(1+C)$ from the cell's usual value at that hour, which mirrors the STL residual that defines an episode.

Normalization. $PM_{2.5}$ and rain are log-transformed; every channel is standardized per cell with training-period statistics, with a floor on the standard deviation and clipping at ± 10.

VII. ARCHITECTURES

Table IV lists the five networks. All read the same 20 channels and all output the residual of Section V. Each was ported from its original code; with the original input width and the same weights, each port reproduces the original output exactly (for ConvLSTM, with its season input, which these data lack, set to identity). A bug in the original patch AFNO, which dropped the imaginary half of every filtered spectrum, was fixed. Fig. 1 shows the two finalists.

AFNO block. Let $\hat{\mathbf{z}} = \text{FFT}_2(\mathbf{z})$ be the orthonormal 2-D real Fourier transform of a feature map whose channels are split into $b = 8$ blocks. At every Fourier mode, each block passes the same small complex MLP, then a soft threshold S_λ:

$$\mathcal{A}(\mathbf{z}) = \mathbf{z} + \text{FFT}_2^{-1}\Bigl(S_\lambda\bigl(W_2\,\text{ReLU}(W_1\hat{\mathbf{z}} + \mathbf{b}_1) + \mathbf{b}_2\bigr)\Bigr), \tag{9}$$

with $\lambda = 0.01$ and the ReLU applied to the real and imaginary parts separately [8]. An AFNO block computes $\mathbf{x} \leftarrow \mathbf{x} + \mathcal{A}(\text{GN}(\mathbf{x}))$, then $\mathbf{x} \leftarrow \mathbf{x} + \text{MLP}(\text{GN}(\mathbf{x}))$ with a 1×1

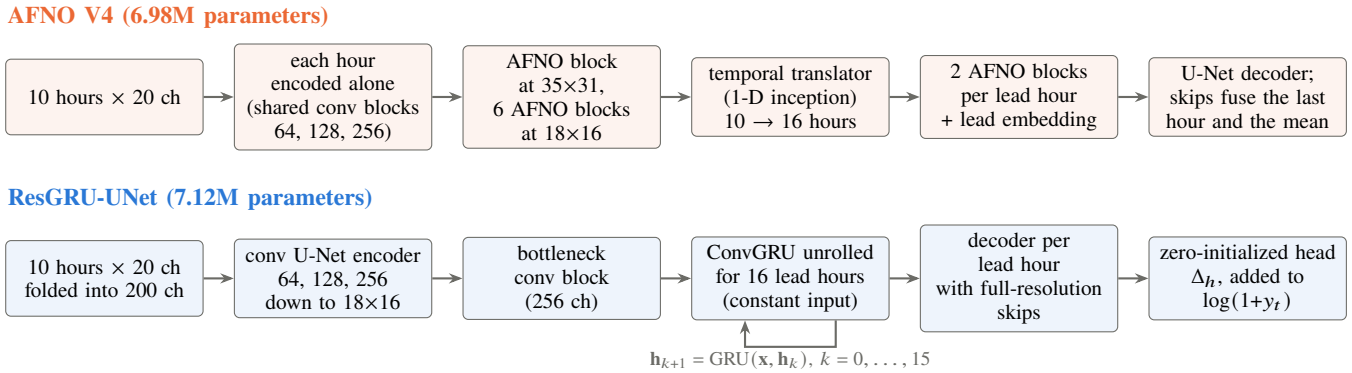


Fig. 1. The two finalists. AFNO V4 encodes each input hour separately and mixes the sequence in a coarse spectral bottleneck; its skip connections keep only the last hour and the mean of the 10 hours. ResGRU-UNet stacks all hours as channels in its first convolution and steps a convolutional GRU through the 16 lead hours. Both output the change Δ_h from persistence.

MLP of width $2d$ and GroupNorm GN. Every block sees the whole map, and the FFT treats opposite edges as neighbours.

ConvGRU. The bottleneck feature $\mathbf{x}$ is fed at every lead hour k and a hidden state $\mathbf{h}_k$ is updated with 3×3 convolutions $*$ [13]:

$$[\mathbf{r}, \mathbf{u}] = \sigma\big(W_g * [\mathbf{x}, \mathbf{h}_k]\big), \tag{10}$$

$$\tilde{\mathbf{h}} = \tanh\big(W_c * [\mathbf{x}, \mathbf{r} \odot \mathbf{h}_k]\big), \tag{11}$$

$$\mathbf{h}_{k+1} = (1 - \mathbf{u}) \odot \mathbf{h}_k + \mathbf{u} \odot \tilde{\mathbf{h}}, \tag{12}$$

starting from $\mathbf{h}_0 = \mathbf{x}$, with GroupNorm inside each gate. The recurrence runs over lead time, not over the input hours: the inputs enter once, through the stacked channels, and each $\mathbf{h}_k$ is decoded with the encoder's skips into the change for lead hour k.

Other networks. FMG-WACLU is a ConvLSTM U-Net that adds wind speed, direction and divergence and gates its features with weather rolled forward by a second ConvLSTM. ConvLSTM reads the 10 input hours recurrently at full resolution, adds an FNO-style spectral block, and shifts the $PM_{2.5}$ map along the wind. Patch AFNO cuts the map into 4×4 patches, mixes them with AFNO blocks and maps each patch back with one linear layer.

VIII. Training and Regularization

Loss. Training minimizes a loss matched to the score,

$$\mathcal{L} = 0.40\,\mathcal{H}_{\log} + 0.25\,G + 0.15\,E + 0.20\,(1 - \rho), \tag{13}$$

where $\mathcal{H}_{\log}$ is the Huber loss on $\log(1 + y)$ with episode cells weighted by 6. Lead hours are weighted linearly from 0.7 to 1.3 in $\mathcal{H}_{\log}$ and G, because errors grow with lead time while the score weights all hours equally. In the loss, G and E add 0.1 to the SMAPE denominator for stable gradients.

Optimization. AdamW [18] with learning rate 2.5×10^{-4} and weight decay 10^{-4}, a one-cycle schedule [19], 16 epochs, effective batch 4, mixed precision and gradient clipping at 1.0. One run takes about 51 minutes (ResGRU-UNet) or 90 minutes (AFNO V4) on one NVIDIA T4.

Regularization. The networks have 3 to 19 million parameters but see only 2,257 training windows from four months, so we use several controls, each for a stated reason:

- channel dropout ($p = 0.1$) in every convolution block of ResGRU-UNet and AFNO V4 and in AFNO V4's MLPs; ConvLSTM uses dropout 0.25 and stochastic depth 0.15;
- decoupled weight decay 10^{-4};
- an exponential moving average of the weights (decay 0.999) [21], which is the model we evaluate;
- selection of the best epoch by the validation score, which acts as early stopping;
- gradient clipping, per-cell standardization with floors, and input clipping, against rare extreme values;
- averaging seeds [22], which removes part of the variance between runs.

Even so, the training curves in Section XII show AFNO V4 fitting the training months more closely than ResGRU-UNet and doing worse on the held-out months.

Calibration. After ensembling, each lead hour is multiplied by a factor c_h fitted on validation from a 21-point grid in $[0.90, 1.10]$ to maximize the two SMAPE terms. The fitted factors rise with lead time from 1.00 to 1.08–1.10, as the episode argument in Section V predicts. We also fit them on three months and test on the fourth (leave one month out), to report calibration without using the scored month.

IX. How the Pipeline Was Developed

The pipeline of Sections IV to VIII was not designed in one go. It grew over seven iterations, each started by a problem that the previous one exposed; the controlled study is iteration 8. Table V lists each move and its reason, and Fig. 2 shows the test score. Iterations 1 to 4 were chosen on a random validation split, which leaked (Section IV), and on feedback from part of the test set, so only their test scores are comparable.

1 to 2: from patch AFNO to ResGRU-UNet. The first models followed FourCastNet: AFNO on 4×4 patches, trained on a random split. They stalled below 0.83 on the test set. The team read this as the global spectral layers acting as low-pass filters that blurred sharp spikes, and moved to a U-Net

TABLE V
DEVELOPMENT ITERATIONS: WHAT CHANGED, WHY, AND THE HELD-OUT TEST SCORE

#	Change	Why	Test
1	AFNO on 4×4 patches, random split	first model: spectral mixing as in global weather models [9]	0.8218
2	U-Net + ConvGRU; output added to the last hour; log Huber loss	AFNO stalled below 0.83; keep local detail and sharp peaks	0.8506
3	fine-tunes: 20 times the loss on missed episode peaks, then pinball loss ($q = 0.75$)	peaks were under-predicted	0.8594
4	blends with models trained at $q = 0.85$ and 0.90	a stronger upward shift	0.8648
5	temporal hold-out; AFNO with a SimVP translator, blended with ResGRU-UNet	the random split leaked	0.8561
6	V3: official STL episodes, exact score for selection, 12-h gap, hour estimation	validation must measure what the test measures	0.8609
7	V4: physics channels, multiscale block, lead-hour weights, 2 seeds, calibration	V3's diagnosis: long-lead errors and a global-episode trade-off	0.8833
8	ResGRU-UNet in the V4 pipeline, 5 seeds, calibration	compare networks with the pipeline fixed	**0.8914**

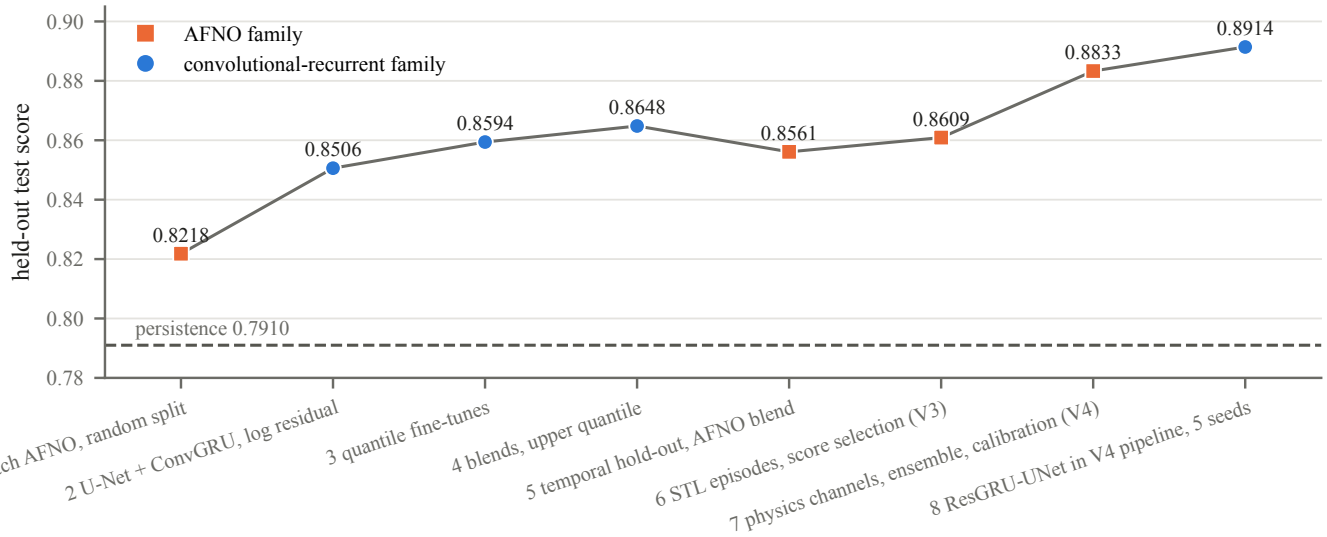


Fig. 2. Held-out test score at each development iteration. The dashed line is persistence (0.7910). Iteration 5 rebuilt the pipeline on an honest split and scored below iteration 4; the rebuilt pipeline overtakes it from iteration 6.

with full-resolution skips for local detail, a ConvGRU, and an output added to the last observed hour. The score rose to 0.8506. Later checks showed that the stated reason was not the mechanism. The patch model's spectral step applied the same complex matrix at every frequency, so apart from its soft threshold it was almost linear and local; a bug also discarded the imaginary half of every filtered spectrum; and each 4×4 patch was rebuilt by one linear layer with no skips. Three things had changed at once (the persistence residual, the skips and a log-space loss), and persistence alone is already strong.

2 to 4: shifting forecasts up. The base ResGRU-UNet under-predicted peaks. The team attributed this to losses that pull forecasts toward the centre and fine-tuned it in steps: a loss 20 times larger on under-predicted episode cells, then a pinball loss at the 0.75 quantile [31], then blends with models trained at the 0.85 and 0.90 quantiles. The test score rose at every step, from 0.8506 to 0.8648. The stated reason was wrong, since SMAPE is symmetric in the log ratio (Section V), but the action was right for another one: episode cells are picked where the truth turned out high, so upward-shifted forecasts score better on the episode terms. Scaling forecasts by hand did not help. "Physics multipliers" that raised stagnant cells by 25% and all other cells by 12% lost 0.006, because the 98% of background cells pay for the shift, and a geometric blend with an AFNO member lost 0.008. Grafting four physics channels onto trained weights failed with NaN values.

4 to 5: an honest split. The random split gave an episode correlation near 0.98 for models that scored about 0.86 on the test set: with overlapping windows, it puts near-copies of every validation window into training. Iteration 5 held out the last 15 to 20% of each month and rebuilt the model as AFNO with a SimVP-style temporal translator [16]. Its best test score, 0.8561, came from a blend with ResGRU-UNet and was below iteration 4. An honest split does not raise the score; it makes choices trustworthy. And choices were still made on the wrong target: a proxy episode mask that marked 0.30% of validation cells against about 2% for the official one, checkpoints chosen on the validation loss, and no gap between

TABLE VI
THE SAME NETWORK IN ITS OWN PIPELINE AND IN THE V4 PIPELINE (HELD-OUT TEST; V4 PIPELINE: SEED 42, NO ENSEMBLE OR CALIBRATION). EACH GAIN BUNDLES EVERY PIPELINE CHANGE, AND FOR PATCH AFNO ALSO ITS BUG FIX

Network	Own pipeline	V4 pipeline	Gain
ResGRU-UNet	0.8506	0.8798	+0.0292
FMG-WACLU	0.8465	0.8710	+0.0245
patch AFNO	0.8218	0.8530	+0.0312

training and validation windows.

5 to 6: validation equal to the metric (V3). V3 rewrote the pipeline so that validation measures exactly what the test measures: the official robust STL episodes, the exact score for choosing checkpoints, a 12-hour gap, per-cell normalization, a lead-hour embedding and weight averaging. It also fixed a hidden mismatch: the ResGRU-UNet models had been trained with the true hour of day but given hours 0 to 9 for every test window. In V3, training keeps the true hour, while validation and test windows get the estimated hour of Section VI, so validation sees the same hour noise as the test. As one model with no blend, V3 reached 0.8609. Its validation now exposed what no earlier pipeline could see: global SMAPE kept improving after epoch 4 while episode SMAPE got worse.

6 to 7: physics and calibration (V4). V4 kept V3's pipeline and added what that diagnosis asked for: the ventilation, advection, tendency and diurnal-anomaly channels of Section VI, one AFNO block at 35×31 for a middle scale [10], lead-hour weights because errors grow with lead time, two seeds, and per-hour calibration fitted on the honest hold-out. Calibration did in one fitted step what iterations 3 and 4 had done by hand, and the physics channels that failed as a graft worked as inputs trained from the start. The test score rose by 0.0224 to 0.8833.

7 to 8: one pipeline for every network. V4 was the best model, but its pipeline had changed together with its network: split, masks, inputs, loss, hour handling and post-processing. The other networks had been trained in older, leakier pipelines, so their scores could not be compared with V4's, and the question "which network is better?" had no clean answer. We therefore moved every network into V4's pipeline. Each was ported unchanged except for its input width (20 channels); ConvLSTM and patch AFNO also received the residual output with a zero-initialized last layer, ConvLSTM lost its season input, and patch AFNO got its bug fix. Each port was checked against its original (Section VII). Each round-one notebook first re-evaluated the original AFNO V4 checkpoint on its own validation set and would have stopped unless it reproduced the logged 0.9110; it did, to five decimals. The windows, masks, loss, recipe and seed are then the same for every network, so the network is the only difference. Table VI shows what the pipeline alone did for the three networks whose original runs are known: each gained 0.025 to 0.031 on the test set.

TABLE VII
ROUND ONE: SEED 42, SAME PIPELINE, NO ENSEMBLE OR CALIBRATION

Network	Params	Validation	Test
ResGRU-UNet	7.12M	**0.9158**	**0.8798**
AFNO V4	6.98M	0.9110	0.8762
FMG-WACLU	3.45M	0.9099	0.8710
patch AFNO	5.71M	0.9031	0.8530
ConvLSTM	19.0M	0.8945	0.8452

TABLE VIII
FIVE SEEDS PER FINALIST. CALIBRATION IS FITTED ON VALIDATION

	AFNO V4	ResGRU-UNet	Gap
Single seeds, val: mean	0.91146	0.91630	+0.00484
standard deviation	0.00059	0.00104	
5-seed mean, val	0.91454	0.91933	+0.00479
+ calibration (one month out)	0.91492	0.92097	+0.00605
Single seeds, test: mean	0.87540	0.87994	+0.00454
standard deviation	0.00130	0.00085	
5-seed mean, test	0.8793	0.8838	+0.0045
+ calibration, test	0.8848	**0.8914**	+0.0066

X. CONTROLLED BACKBONE STUDY

The study fixes everything in Sections IV to VIII and changes only the network. It ran in three stages: one seed for all five networks; the two finalists with two seeds and the full V4 recipe; then five seeds per finalist.

Round one. Table VII. ResGRU-UNet leads, and validation and test order the five networks the same way, which supports the validation design. Under the rule, ResGRU-UNet is better than AFNO V4 (+0.0048 validation, +0.0036 test), FMG-WACLU ties with it, and patch AFNO and ConvLSTM are worse.

Two and five seeds. With two seeds, ensembles and calibration for both, ResGRU-UNet again leads: 0.9175 against 0.9131 on validation and 0.8898 against 0.8833 on test. Five seeds give Table VIII and Fig. 3. The validation gap between the five-seed means is +0.00479, with a 95% block-bootstrap interval of $[+0.00383, +0.00578]$ (blocks of 13 consecutive windows within each month, 2,000 resamples). Every ResGRU-UNet seed beats every AFNO V4 seed on validation and on test (25 of 25 pairs); its worst seed (0.8788 test) beats AFNO V4's best (0.8767).

Pipeline against backbone. Moving ResGRU-UNet from iteration 2 into the V4 pipeline raised its test score from 0.8506 to 0.8798 (+0.0292, one seed, no calibration). At the same setting the gap to AFNO V4 is +0.0036, or +0.0045 averaged over five seeds. The pipeline therefore moved the score 6 to 8 times more than the choice between the two best networks (4.4 times against the full-recipe gap of +0.0066). The pipeline gain bundles several changes that we did not separate.

XI. RESULTS

A. Regression metrics

Table IX gives MAE, RMSE, R^2 and skill on validation for every model. The ridge baseline is one linear model shared by all cells: from the 20 channels at the last hour and $\log(1+y)$

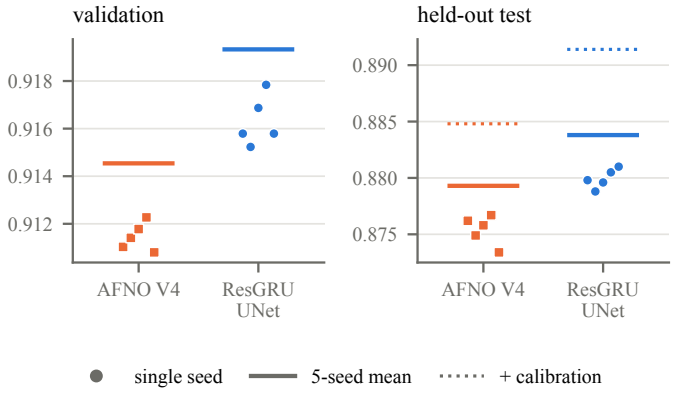


Fig. 3. Every seed of each finalist. The two families do not overlap on validation or on the held-out test.

TABLE IX
VALIDATION REGRESSION METRICS (MAE, RMSE IN $\mu g\,m^{-3}$) OVER ALL 5.9×10^7 VALUES, WITH THE BENCHMARK SCORE S. SKILL: LOWER RMSE THAN PERSISTENCE

Model	MAE	RMSE	R^2	Skill	S
Persistence	10.44	26.46	0.767	0.0%	0.8515
Seasonal naive (24 h)	11.53	26.80	0.761	−1.3%	0.7907
Ridge regression	8.51	21.48	0.847	18.8%	0.8637
Round one, seed 42					
patch AFNO	6.85	19.02	0.880	28.1%	0.9031
ConvLSTM	7.05	18.93	0.881	28.5%	0.8945
FMG-WACLU	7.14	19.32	0.876	27.0%	0.9099
AFNO V4	7.31	20.17	0.865	23.8%	0.9110
ResGRU-UNet	6.57	18.57	0.885	29.8%	0.9158
Five-seed means					
AFNO V4	6.99	19.45	0.874	26.5%	0.9145
+ calibration	7.76	20.90	0.855	21.0%	0.9149
ResGRU-UNet	**6.32**	**18.04**	**0.892**	**31.8%**	0.9193
+ calibration	7.12	19.71	0.871	25.5%	**0.9210**

at the nine earlier hours (30 features with a bias), it predicts the 16 log changes, fitted in closed form on 2.45 million training cell-windows (every fourth cell in each direction) with the penalty chosen on validation. With zero weights it is persistence. Every network beats all three baselines on every metric. The ResGRU-UNet ensemble has 31.8% lower RMSE and 39.4% lower MAE than persistence. Ridge improves on persistence on validation (score 0.8637) but barely on the test set (0.7951 against 0.7910).

B. By lead hour and metric term

Fig. 4 shows the error growing with lead time: ResGRU-UNet's R^2 falls from 0.992 at hour 1 to 0.830 at hour 16, and persistence's from 0.985 to 0.674. Fig. 5 splits the score into its three terms. ResGRU-UNet is better on all three at every hour; the score gap grows from +0.0018 at hour 1 to about +0.006 at hours 10 to 15. Of the five-seed gap, 56% comes from global SMAPE, 23% from episode SMAPE and 22% from episode correlation.

C. Rankings depend on the metric

ResGRU-UNet is first on all six measures in Fig. 6, but the other four reorder. AFNO V4 is second on the benchmark score yet last on MAE, RMSE and R^2; ConvLSTM is last on the score yet second on RMSE and R^2. The score is relative and focused on episodes, while raw-scale MAE, RMSE and R^2 are dominated by high concentrations. One metric alone would have misjudged the weaker networks.

D. Calibration trades error for score

Calibration raises the score but worsens MAE, RMSE and R^2 for both finalists (Fig. 7): the ResGRU-UNet ensemble moves from MAE 6.32 to 7.12 while its score rises from 0.9193 to 0.9210. The factors push forecasts up, which the episode terms reward and absolute and squared errors penalize. The uncalibrated mean is the better model for MAE, RMSE and R^2; the calibrated one is better for the benchmark score.

E. Where ResGRU-UNet wins

It is ahead in all four validation months and in five of six concentration bins (Fig. 8). Both models under-predict episode cells (mean log bias −0.160 for AFNO V4 and −0.157 for ResGRU-UNet) and over-predict the other cells (+0.084 and +0.062), most of all in the cleanest bin (Fig. 8, middle). ResGRU-UNet's smaller over-prediction outside episodes is consistent with its larger gain in the global term.

F. Spatial structure

Fig. 9 maps the time-averaged errors. $PM_{2.5}$ and the errors of persistence concentrate over the Indo-Gangetic Plain and the eastern coal belt; ResGRU-UNet removes much of that error. It has lower MAE than AFNO V4 in 90.5% of cells and lower SMAPE in 94.7% (Fig. 10), with the largest gains over the Indo-Gangetic Plain and central India. The median per-cell R^2 over time is 0.872 for ResGRU-UNet, 0.846 for AFNO V4 and 0.694 for persistence; it is lowest north of 32°N (median 0.764), where the air is cleanest (13.8 against 42.3 $\mu g\,m^{-3}$ south of it). Fig. 11 shows the window with the most episode cells.

G. Cost

ResGRU-UNet is also cheaper: 25.8 against 48.6 ms per window at inference, 0.23 against 1.36 GB of peak inference memory at batch 4, and 191 against 338 s per training epoch on a T4.

XII. Ablation studies

What V4's pipeline steps added. V4's test gain over V3 (+0.0224) splits into +0.0147 from the per-seed changes (physics channels, multiscale block, horizon weights), +0.0024 from the two-seed ensemble and +0.0054 from calibration (65%, 10% and 24%), measured by submitting the intermediate forecasts.

Why the convolutional-recurrent network wins. We tested four hypotheses, each with one change to one finalist and one seed (Table X), with predictions written down first. A

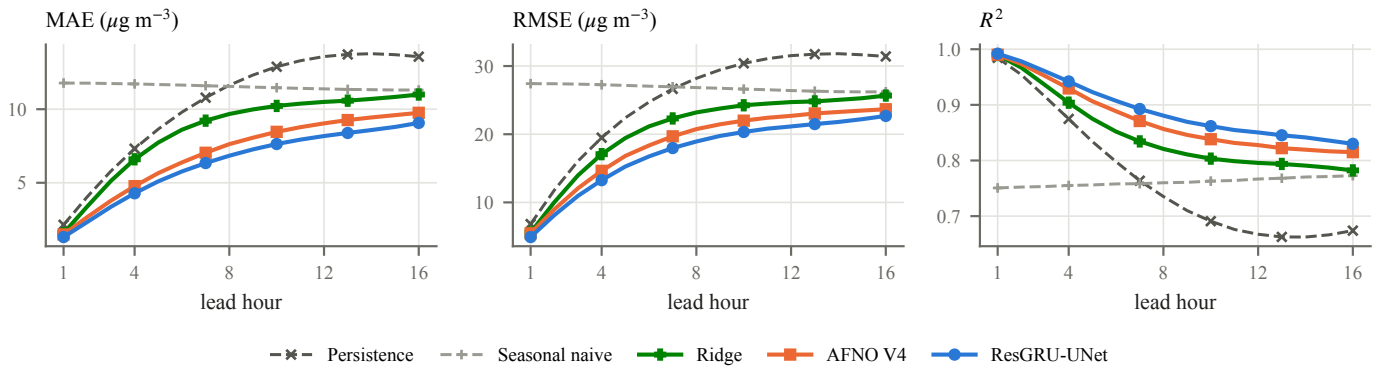


Fig. 4. Validation MAE, RMSE and R^2 by lead hour (five-seed means, no calibration). ResGRU-UNet is best at every hour. Seasonal naive overtakes persistence from hour 8, when the daily cycle outweighs the last observed state.

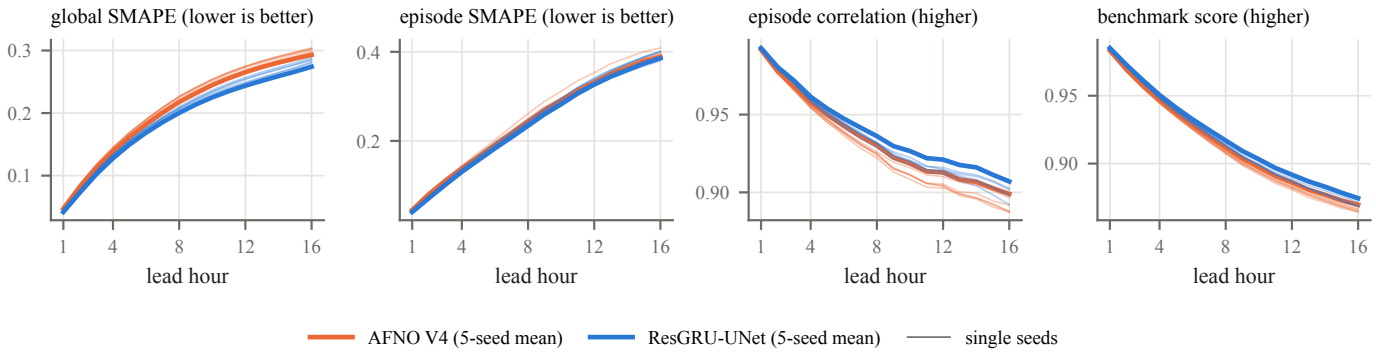


Fig. 5. The three terms of the benchmark score and the score itself by lead hour, on validation. Thin lines: single seeds; thick lines: five-seed means.

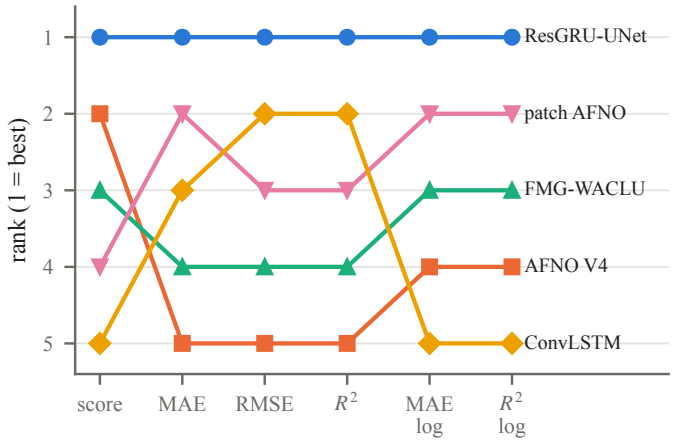


Fig. 6. Round-one ranks (seed 42) under six measures. Only the winner is stable.

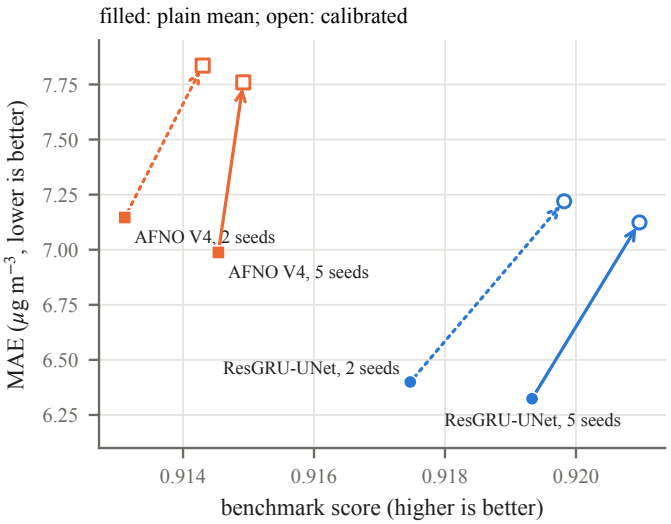


Fig. 7. Calibration moves both finalists to a higher score and a higher MAE (validation; in sample for 2 seeds, leave one month out for 5 seeds).

change counts only if it moves validation by more than $\max(2\,\mathrm{sd}, 0.002)$ from its family's five-seed mean:

- H1: AFNO V4's skips keep only the last hour and the mean;
- H2: the FFT wraps opposite edges of a non-periodic map;
- H3: the soft threshold removes small details;
- H4: ResGRU-UNet wins because of its recurrence over lead time.

None passes the threshold at one seed. H1 (skips that fuse all 10 hours, started equal to V4's fusion) moves AFNO V4 about a quarter of the way toward ResGRU-UNet. Replacing ResGRU-UNet's ConvGRU with per-lead convolution blocks of the same size costs about a third of the gap, yet it still

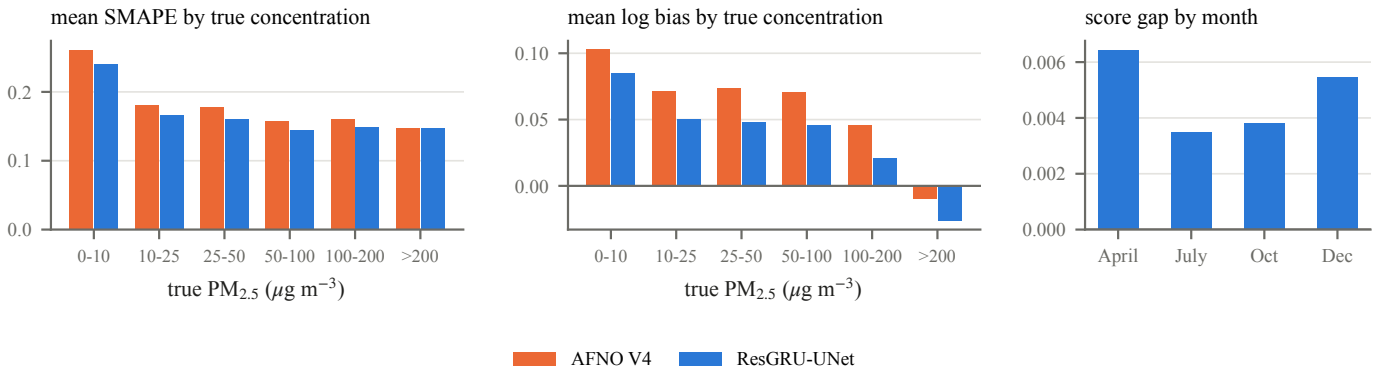


Fig. 8. Five-seed means on validation. Left: mean SMAPE by true concentration. Middle: mean log bias (positive: too high). Right: score gap by month (ResGRU-UNet minus AFNO V4).

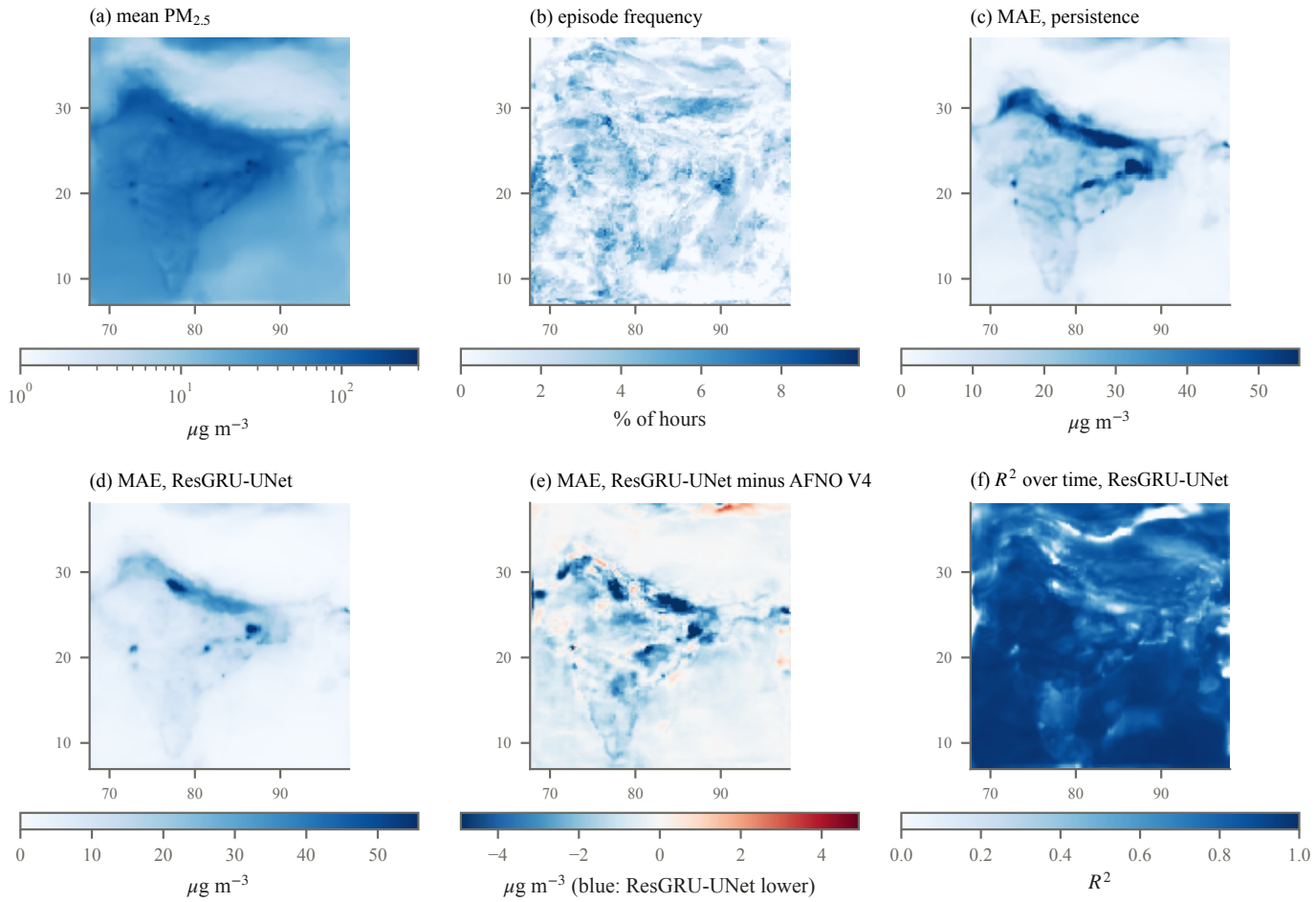


Fig. 9. Time-averaged validation maps. (a) Mean $PM_{2.5}$. (b) Share of hours that are episodes. (c, d) MAE of persistence and of the ResGRU-UNet five-seed mean. (e) MAE difference, ResGRU-UNet minus AFNO V4. (f) R^2 of each cell's time series, ResGRU-UNet.

beats every AFNO V4 seed: the recurrence is not needed to win, but its share is unresolved. Padding also strengthens the soft threshold, because the orthonormal FFT spreads the same energy over four times as many coefficients; H3 suggests this matters little.

A generalization gap. The training curves (Fig. 12) show a pattern none of the AFNO changes altered. Every AFNO run reaches a lower training loss than ResGRU-UNet but a higher validation loss: at a training loss of 0.09, its validation loss is 6.8% higher on average. Its score peaks at epoch 4 or 5 and then falls by 0.009 to 0.010; ResGRU-UNet peaks at epoch 6 or 7 and falls by 0.003 to 0.005. Without its recurrence, ResGRU-

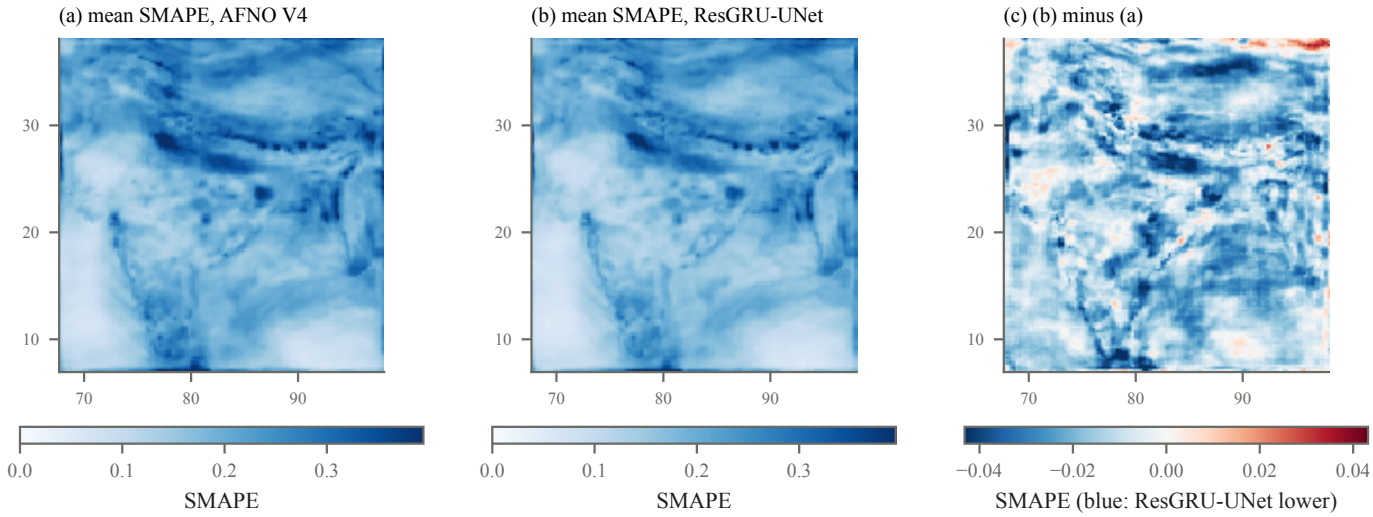


Fig. 10. Mean SMAPE per cell for the five-seed means of AFNO V4 (a) and ResGRU-UNet (b), and their difference (c). ResGRU-UNet is lower in 94.7% of cells.

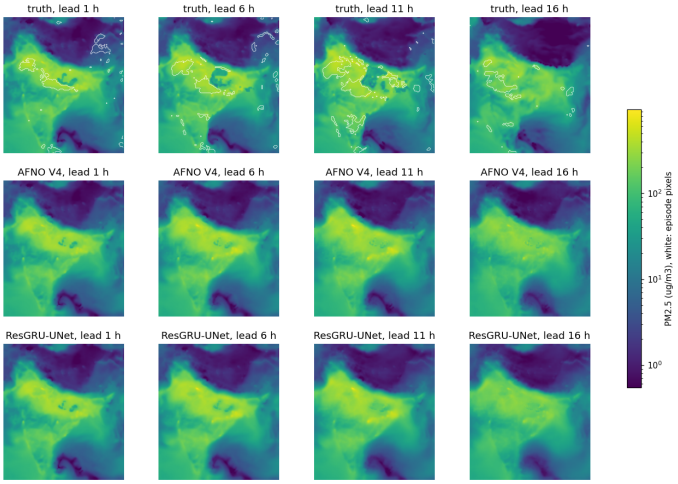


Fig. 11. The validation window with the most episode cells (December): truth with episode cells outlined (top), AFNO V4 (middle) and ResGRU-UNet (bottom) at lead hours 1, 6, 11 and 16, in grid coordinates on a log colour scale. Both smooth the episode structure; ResGRU-UNet scores higher on this window (0.9116 against 0.9103) but has the lower episode correlation (0.928 against 0.944).

TABLE X

SINGLE-CHANGE ABLATIONS (SEED 42). d: VALIDATION CHANGE FROM THE FAMILY'S FIVE-SEED MEAN; d/G: SHARE OF THE 0.00484 GAP. THRESHOLD 0.002

	Change	d	d/G	Test
H1	AFNO V4, skips fuse all 10 hours	+0.00113	+0.23	0.8748
H2	AFNO V4, zero-padded FFT	−0.00084	−0.17	0.8745
H3	AFNO V4, no soft threshold	−0.00052	−0.11	0.8758
H4	ResGRU-UNet, ConvGRU replaced	−0.00174	−0.36	0.8798

UNet moves about 41% of the way toward AFNO's validation loss. These observations are exploratory, but they suggest AFNO V4 fits the training months more and transfers less to

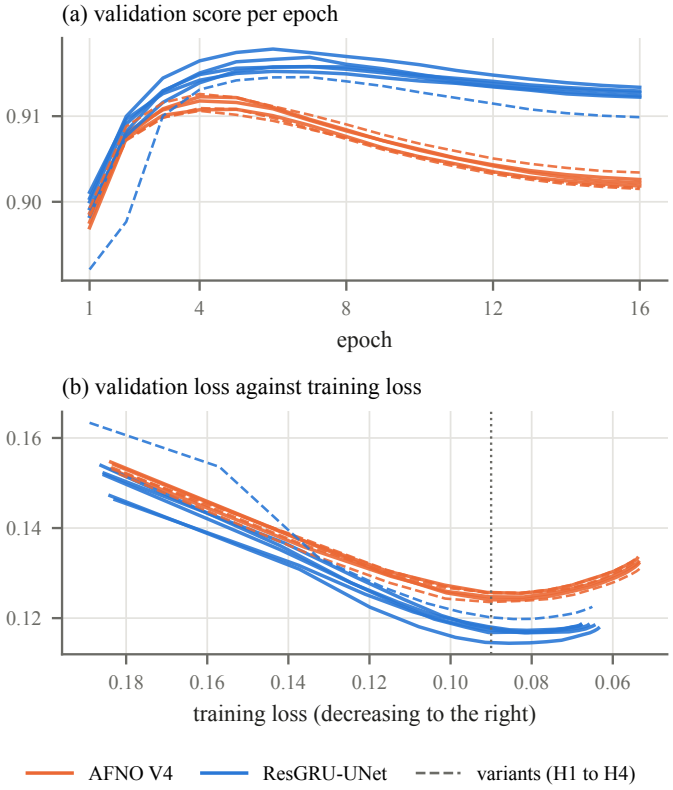


Fig. 12. Training curves of the 12 logged runs: (a) validation score per epoch; (b) validation loss against training loss, with a dotted line at a training loss of 0.09.

new months. Its global, wrap-around spectral mixing, per-hour encoder and translator remain candidates for the cause.

XIII. LIMITATIONS

- The data are simulations, not measurements, and skill on WRF-Chem fields need not carry over to monitored

$PM_{2.5}$.

- There is one validation period and one test set. Validation picked each run's best epoch and fitted the calibration, so validation scores are optimistic; the test set is the only untouched measure, and it reports only the benchmark score.
- The score's weights are not disclosed; we assume equal weights on validation.
- The recipe was tuned for AFNO V4, which if anything favours it; ResGRU-UNet was not tuned.
- The later rounds were designed after round one's result, adjacent bootstrap blocks overlap in time, and the ablations use one seed each.
- The pipeline gain bundles several changes that we did not separate.

XIV. Conclusion

For 16-hour $PM_{2.5}$ forecasting over India, the pipeline mattered more than the network. Physics channels for mixing and transport, a log-space persistence residual, validation that reproduces the test, a loss matched to the score, and ensembling with calibration gave the same network a gain 6 to 8 times the gap between the two best networks. Inside one fixed pipeline, a plain convolutional-recurrent U-Net beat the Fourier neural operator hybrid on every seed pair, lead hour, month and regression metric, and it is also cheaper to run. No single component we tested explains the gap; the Fourier model generalizes less well across months, which fits a transport problem that is local and not periodic. A natural next step is a hybrid that adds a gradient-boosted correction, built on the Phase 1 features, to the network's forecast.

Reproducibility

The code is at https://github.com/Vetri-78640/dl_pm2.5: the shared pipeline and every network (`src/`), the notebook builders and Kaggle tools (`scripts/`), the dry runs, port check and config test (`tests/`), six local notebooks with outputs (`notebooks/`; `notebooks/kaggle/` holds the Kaggle notebooks exactly as they ran, with their run logs), the study settings (`configs/`), the run records (`results/`), the AFNO V4 script and the original notebooks of the ported networks (`reference/`), and the result tables, figures and this report (`reports/`). The competition data is not included; see `data/README.md`.

Team contributions

- **Vetriselvan R**: study design and decision rules (fixed before training); model architectures: AFNO V4, ResGRU-UNet, the ports of the other three networks, the port check and the ablation variants; the round-one, final-round and ablation runs; and this report.
- **Hrishi Dongre**: training pipeline: loss, optimization and schedule, seed runs and the Kaggle notebooks.
- **Pampana Sree Adithya**: experiments and analysis: 5-seed statistics and block bootstrap, the ablation verdict, figures and error maps.
- **Ishan Goyal**: data and pre-processing: exploration, physics channels, normalization, the temporal split and episode masks.
- **Vedant Madne**: evaluation and post-processing: the metric, calibration, ensembles, the baselines including ridge, and the regression metrics.

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